Weighing an Explosion

用尺子和时钟称量一场爆炸：Taylor 的爆炸波估算

改编自 G. I. Taylor, *Proc. R. Soc. A* 201 (1950)　难度 ★　物理考点：量纲分析、流体力学、激波

本单元目标

- **语言**：掌握表达量级与标度的句式（*scales as, of the order of, negligible compared with*），能流利读出含指数和比例关系的公式。
- **词汇**：掌握约 50 个词条，覆盖通用学术词、专业术语和进阶词。
- **数理**：用量纲分析导出 $R \propto t^{2/5}$，由真实数据估算爆炸能量，并逐一检验模型的假设。
- **学术能力**：学会区分标度论证、完整理论和实验数据各自能说明什么，写出第一篇英文摘要。

学习安排（共约 3 小时 40 分钟，其中课内 2 小时）

阶段	内容	时间	要求
课前	Main Text 通读，Vocabulary 预习	40 min	带着 Reading Check 的问题读，遇到公式先读出声
课内	Warm-up 与预习检查	10 min	口头讨论
	精读难点段落（¶4、¶6、¶8），Reading Check	25 min	说清论证结构，不逐句翻译
	Language Focus	10 min	朗读公式，改错和翻译
	Work It Out 与 Code Lab	60 min	用英文写关键步骤，可用计算器
	Output A：口头讲解	15 min	2 分钟英文讲解
课后	Competition Corner，Output B，词汇复习	约 1 h	书面完成

Warm-up 热身

课内·10 min

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1. **A delayed wave.** Enrico Fermi watched the first atomic test from about ten miles (16 km) away. Estimate how long the blast took to reach him. Why might your estimate be slightly too long?
2. **A sense of scale.** One kiloton of TNT is defined as 4.184×10^{12} J. Assume that Shanghai uses about 2×10^{11} kWh of electricity per year. For how long could the energy of a 17-kiloton explosion supply the city?
3. **A thought experiment.** You are given a photograph of an expanding fireball, with a time stamp and a scale bar, but no other information. Which physical quantities would you want to know in order to estimate the energy of the explosion? Which do you think you could ignore?

Main Text 正文

课前·40 min

Weighing an Explosion with a Ruler and a Clock

Abstract. In 1950 G. I. Taylor estimated the energy of the first atomic explosion using a series of published photographs. This article reconstructs his argument. We show that dimensional analysis fixes how the radius of a strong blast wave grows with time, $R \propto t^{2/5}$, and that the data follow this law closely. A full fluid-dynamical calculation supplies the remaining constant, which turns out to be close to unity. The resulting energy, about 17 kilotons of TNT, is of the same order as modern estimates. The case shows how much a simple scaling argument can reveal, provided that its assumptions are examined with care.

1 Introduction

1 At 05:29 on 16 July 1945, the first nuclear device was detonated in the New Mexico desert. High-speed cameras recorded the growth of the fireball, with successive frames a fraction of a millisecond apart. Two years later, some of these records were declassified. Each frame carried a time stamp and a distance scale, but there was no statement of the energy released. To most readers the pictures were merely dramatic. To Sir Geoffrey Taylor, a Cambridge fluid dynamicist, they were data.

detonate 起爆
fireball 火球
declassify 解密
time stamp 时间标记
merely 仅仅
fluid dynamicist 流体力学家

2 Taylor was unusually well prepared. In 1941 the British government had asked him what a very intense explosion would do to the surrounding air, and he had predicted how the resulting blast wave should grow. His report remained classified until 1949. In 1950 he published the theory, together with a comparison with the photographs and an estimate of the energy. According to his biographer, the US Army mildly rebuked him for doing so. This article reconstructs the central argument.

blast wave 爆炸波
classified 保密的
rebuke 斥责

2 The scaling argument

3 Suppose that a large energy E is released almost instantaneously at a point in still air of density ρ. A spherical shock front rushes outward, sweeping up the air in its path (Fig. 1a). We make two assumptions. First, the source is effectively a point: the mass of air swept up far exceeds the mass of the device, whose size and composition are soon forgotten. Second, the shock is strong: the pressure just behind the front is so much greater than the ambient pressure that the latter is negligible.

shock front 激波波阵面
sweep up 扫起、卷起
ambient 周围环境的
negligible 可忽略的

4 Under these assumptions, only three quantities can determine the radius R of the front: E, ρ and the time t since detonation. Their dimensions are $[E] = \mathrm{M\,L^2\,T^{-2}}$, $[\rho] = \mathrm{M\,L^{-3}}$ and $[t] = \mathrm{T}$, and only one combination of them has the dimensions of length. Therefore

$$R = S\left(\frac{E}{\rho}\right)^{1/5} t^{2/5}, \tag{1}$$

where S is a dimensionless constant that may depend on γ, the ratio of specific heats of air, but not on E, ρ or t. Without solving a single differential equation, we have obtained the exponent of time.

dimensionless 无量纲的
ratio of specific heats 比热比
exponent 指数

5 The same reasoning implies more. Since the problem contains no other length or time scale, the flow must look identical at every moment, provided that distances are mea-

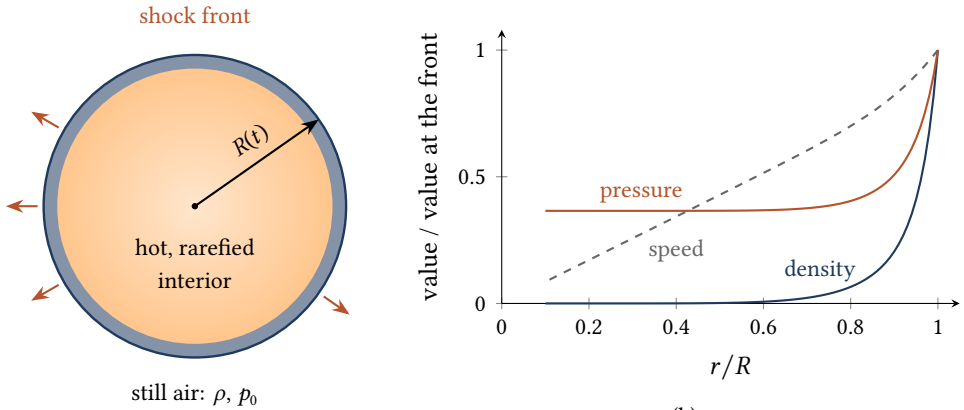


Figure 1. (a) A spherical blast wave. (b) The self-similar profile for $\gamma = 1.4$, computed from the equations of compressible flow. About 80% of the swept-up air lies in the outer 10% of the radius.

Table 1. Radius of the blast wave at time t after detonation (Taylor 1950).

t (ms)	R (m)	t (ms)	R (m)	t (ms)	R (m)
0.10	11.1	1.08	38.9	3.80	62.9
0.24	19.9	1.22	41.0	4.07	64.3
0.38	25.4	1.36	42.8	4.34	65.6
0.52	28.8	1.50	44.4	4.61	67.3
0.66	31.9	1.65	46.0	15.0	106.5
0.80	34.2	1.79	46.9	25.0	130.0
0.94	36.3	1.93	48.7	34.0	145.0
		3.26	59.0	53.0	175.0
		3.53	61.1	62.0	185.0

sured in units of $R(t)$. Such a flow is called self-similar. Figure 1b shows its profile: most of the air piles up in a thin, dense shell just behind the front, while the interior is hot, rarefied and nearly uniform in pressure.

self-similar 自相似的
profile 分布曲线
pile up 堆积
rarefied 稀薄的

6 Dimensional analysis cannot yield the value of S, however, and here popular accounts often mislead. Taylor did not stop at a scaling argument. He substituted a self-similar form into the equations of motion, continuity and state for a compressible gas, and integrated them step by step. Rearranging Eq. (1) gives

mislead 误导
continuity 连续性（方程）
compressible 可压缩的

$$E = K\,\frac{\rho R^5}{t^2}, \qquad K = S^{-5}, \tag{2}$$

and for $\gamma = 1.4$ he found $K = 0.856$, so that $S \approx 1.03$, remarkably close to unity.

unity （数学）1

3 Comparison with the photographs

7 Table 1 lists the radii that Taylor measured. Taking logarithms of Eq. (1) gives $\log R = \tfrac{2}{5}\log t + \text{constant}$, so a log–log plot should be a straight line of slope $2/5$. Figure 2 shows that, from $R \approx 20$ m to 185 m, the points follow this line closely over more than two decades in time. Taylor found that $R^5/t^2 \approx 6.67 \times 10^{13}\ \text{m}^5\,\text{s}^{-2}$. With $\rho = 1.25\ \text{kg}\,\text{m}^{-3}$, Eq. (2) then gives $E \approx 7.1 \times 10^{13}$ J, equivalent to about 17 kilotons of TNT.

logarithm 对数
slope 斜率
kiloton 千吨（TNT 当量）

8 How good is this figure? A recent radiochemical analysis puts the total yield at 24.8 ± 2

radiochemical 放射化学的

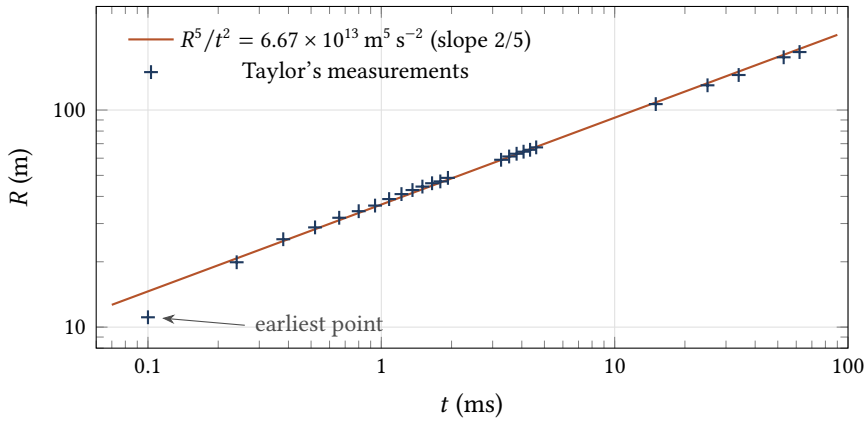


Figure 2. Taylor's measurements on logarithmic axes. Figure redrawn from Table 1.

kilotons, against a long-standing official value of 21. Taylor's result looks low, but two points should be kept in mind. His E counts only the energy that was not radiated away beyond the fireball. It also depends on γ: for $\gamma = 1.3$, a value more appropriate to very hot air, K rises to 1.167 and the estimate to about 23 kilotons. Taylor preferred the lower figure, arguing that radiation and high-temperature effects roughly cancelled each other.

4 Discussion

9 Equation (2) is highly sensitive to its input: because $E \propto R^5$, a 2% error in the radius becomes a 10% error in the energy. Both assumptions must also fail eventually. At the earliest times the device itself and radiation still matter, and the first point in Table 1 does fall below the line. At late times the shock weakens, the ambient pressure can no longer be ignored, and Eq. (1) must break down. The surprise is not that the law has limits, but that it holds so well between them, even though a constant γ is certainly untrue at such temperatures.

10 The same law appears far beyond this case. John von Neumann and Leonid Sedov reached it independently in the 1940s, and today the Sedov–Taylor phase is a standard stage in the evolution of supernova remnants, whose energies are some thirty orders of magnitude larger. Yet the lesson is not that calculation can be dispensed with. A back-of-the-envelope argument tells you what to plot. Only a full theory tells you what the numbers mean, and only the data can tell you whether either is right.

supernova remnant 超新星遗迹
dispense with 免去、省去
back-of-the-envelope 粗略估算的

References

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Vocabulary 词汇

课前预习·课后复习

Word	释义	Collocation / example
A. General academic words 通用学术词		
reconstruct /ˌriːkənˈstrʌkt/	v. 重构、还原	reconstruct an argument
assumption /əˈsʌmpʃn/	n. 假设	make / examine an assumption
determine /dɪˈtɜːmɪn/	v. 决定；测定	determine the radius
imply /ɪmˈplaɪ/	v. 意味着、蕴含	The result implies that …
yield /jiːld/	v. 得出；n. 爆炸当量、产额	yield a value; the total yield
estimate /ˈestɪmət/ (n.) /ˈestɪmeɪt/ (v.)	n./v. 估算	a modern estimate; estimate the energy
exceed /ɪkˈsiːd/	v. 超过	far exceed the mass of the device
effectively /ɪˈfektɪvli/	adv. 实际上、等效地	effectively a point
instantaneously /ˌɪnstənˈteɪniəsli/	adv. 瞬时地	released almost instantaneously
intense /ɪnˈtens/	adj. 强烈的	a very intense explosion
substitute /ˈsʌbstɪtjuːt/	v. 代入	substitute A into B（注意介词 into）
integrate /ˈɪntɪgreɪt/	v. 积分	integrate the equations step by step
rearrange /ˌriːəˈreɪndʒ/	v. 移项、整理	Rearranging Eq. (1) gives …
uniform /ˈjuːnɪfɔːm/	adj. 均匀的	nearly uniform in pressure
appropriate /əˈprəʊpriət/	adj. 合适的	appropriate to very hot air
sensitive /ˈsensətɪv/	adj. 敏感的	highly sensitive to its input
independently /ˌɪndɪˈpendəntli/	adv. 独立地	reach a result independently
phase /feɪz/	n. 阶段	the Sedov–Taylor phase
evolution /ˌiːvəˈluːʃn/	n. 演化	the evolution of supernova remnants
eventually /ɪˈventʃuəli/	adv. 最终	must fail eventually
B. Technical terms 专业术语		
detonate /ˈdetəneɪt/	v. 起爆（n. detonation）	the time since detonation
blast wave /blɑːst weɪv/	n. 爆炸波	a strong blast wave
shock front /ʃɒk frʌnt/	n. 激波波阵面	just behind the shock front
fireball /ˈfaɪəbɔːl/	n. 火球	the growth of the fireball
dimensional analysis /daɪˈmenʃənl əˈnæləsɪs/	n. 量纲分析	Dimensional analysis fixes …
dimensionless /daɪˈmenʃənləs/	adj. 无量纲的	a dimensionless constant
self-similar /ˌself ˈsɪmələ/	adj. 自相似的	a self-similar flow
ambient /ˈæmbiənt/	adj. 周围环境的	ambient pressure
density /ˈdensəti/	n. 密度	air of density ρ
ratio of specific heats /ˈreɪʃiəʊ, spəˈsɪfɪk hiːts/	n. 比热比	the ratio of specific heats of air
compressible /kəmˈpresəbl/	adj. 可压缩的	compressible flow
continuity /ˌkɒntɪˈnjuːəti/	n. 连续性	the equation of continuity
exponent /ɪkˈspəʊnənt/	n. 指数	the exponent of time
logarithm /ˈlɒgərɪðəm/	n. 对数	take logarithms of an equation
slope /sləʊp/	n. 斜率	a straight line of slope 2/5
radius /ˈreɪdiəs/, pl. radii /ˈreɪdiaɪ/	n. 半径	measure the radii
kiloton /ˈkɪlətʌn/	n. 千吨（TNT 当量）	17 kilotons of TNT
radiochemical /ˌreɪdiəʊˈkemɪkl/	adj. 放射化学的	a radiochemical analysis

Word	释义	Collocation / example
supernova remnant /ˌsuːpəˈnəʊvə ˈremnənt/	n. 超新星遗迹	the evolution of a supernova remnant
fluid dynamicist /ˈfluːɪd daɪˈnæmɪsɪst/	n. 流体力学家	a Cambridge fluid dynamicist
C. Advanced words 进阶词		
merely /ˈmɪəli/	adv. 仅仅	merely dramatic
rebuke /rɪˈbjuːk/	v./n. 斥责	mildly rebuke sb. for doing sth.
sweep up /swiːp ʌp/	phr. 扫起、卷起	sweep up the air in its path
pile up /paɪl ʌp/	phr. 堆积	Most of the air piles up behind the front.
rarefied /ˈreərɪfaɪd/	adj. 稀薄的	a hot, rarefied interior
mislead /ˌmɪsˈliːd/	v. 误导（过去式 misled）	Popular accounts often mislead.
unity /ˈjuːnəti/	n. （数学）1	close to unity
dispense with /dɪˈspens wɪð/	phr. 免去、省去	Calculation cannot be dispensed with.
back-of-the-envelope /ˌbæk əv ðiˈenvələʊp/	adj. 粗略估算的	a back-of-the-envelope argument
break down /breɪk daʊn/	phr. （理论）失效	The equation must break down at late times.

Word Work

Complete each sentence with one word or phrase from the list, changing its form where necessary.

negligible *break down* *unity* *exceed* *substitute* *yield* *sensitive* *dispense with*

1. Air resistance is _____ for a heavy ball falling a short distance.
2. The ideal-gas law begins to _____ at very high densities.
3. _____ this expression into Eq. (2) gives the final result.
4. Because the ratio is very close to _____, the two quantities are almost equal.
5. The speed of the electrons _____ half the speed of light.
6. The output is highly _____ to small changes in temperature.
7. Solving this equation numerically _____ a value of about 1.03.
8. With a good symmetry argument, we can _____ most of the algebra.

Reading Check 阅读理解

课内·与精读合计 25 min

A. True or false? Give the paragraph that supports your answer.

1. Taylor developed his theory after he had seen the photographs.
2. Dimensional analysis alone gives both the exponent of t and the value of S.
3. Taylor's value of E includes all the energy released by the explosion.
4. The first point in Table 1 supports the scaling law as well as the others do.

B. Answer in one or two sentences of English.

5. Which physical quantities does each of the two assumptions in ¶3 allow us to remove from the problem?

6. ¶1 ends: "To most readers the pictures were merely dramatic. To Sir Geoffrey Taylor …, they were data." What does the contrast tell us about how a scientist looks at a picture?
7. According to ¶6, what do popular accounts often get wrong about Taylor's work?
8. Why does the author call the success of the law a "surprise" in ¶9?

C. Thinking like a referee.

9. Match each function with a paragraph: (a) states the model's limits; (b) sets out the assumptions; (c) compares the result with an independent measurement; (d) draws a general lesson.
10. The bomb was fired on a tower about 30 m above the ground, and the photographs show the *luminous edge* of the fireball. Name two things that the article assumes without saying, and explain why each one could matter.

Language Focus 语言聚焦：描述量级与标度

课内·10 min

物理英语里，比例和量级有一套固定说法。下面这些句式几乎每篇论文都会用到。

Pattern	Example	注意
A is proportional to B	E is proportional to the fifth power of R.	介词用 to，不用 with
A scales as B	The radius scales as $t^{2/5}$.	后接幂律，也说 scales like
A is of the order of B	The energy is of the order of 10^{14} J.	美式常用 on the order of
n orders of magnitude larger	The energy is thirty orders of magnitude larger.	不能省略 of magnitude
A is negligible compared with B	p_0 is negligible compared with the pressure behind the front.	被忽略的量作主语
A is much greater than B	$p_2 \gg p_0$ reads "p_2 is much greater than p_0".	$\ll$ 读 much less than
to within x	The two values agree to within 30%.	表示误差不超过
close to unity	S is close to unity.	比 close to one 更正式

A. Read aloud.

(1) $R \propto t^{2/5}$ (2) $E = K\rho R^5/t^2$ (3) $p_2 \gg p_0$ (4) $\delta E/E \approx 5\,\delta R/R$ (5) $E \sim 10^{14}$ J

B. Correct the error in each sentence.

1. The radius is proportional with $t^{2/5}$.
2. The pressure behind the front is negligible compared with the ambient pressure, because the shock is strong.
3. The energy of the explosion is the order of 10^{14} J.
4. Since E scales as R^5, doubling the radius makes the energy five times larger.
5. A supernova releases thirty orders more energy than this explosion.

C. Translate into English.

1. 爆炸波的半径随时间的 2/5 次方增长。
2. 激波后的压强约为环境压强的 80 倍，因此后者可以忽略。
3. 这一估计与现代结果的偏差在 30% 左右。
4. 半径上 2% 的误差会使能量产生 10% 的误差。

Work It Out 动手推演

课内·45 min

Work on paper and write the key steps in English. Use $\rho = 1.25\,\mathrm{kg\,m^{-3}}$, ambient pressure $p_0 = 1.01 \times 10^5$ Pa, sound speed $c = 340\,\mathrm{m\,s^{-1}}$ and $1\,\mathrm{kt} = 4.184 \times 10^{12}$ J.

Task 1 Three variables, one length

10 min

(a) Write $R = S\,E^a \rho^b t^c$. Find a, b and c by matching dimensions.

(b) Now keep the ambient pressure p_0 in the problem. Find a dimensionless group Π that contains p_0, E, ρ and t, with p_0 to the first power. Explain why dimensional analysis can now only give $R = (Et^2/\rho)^{1/5} f(\Pi)$, where f is an unknown function.

(c) Setting $\Pi = 1$ gives a characteristic time t^* and radius R^*. Find expressions for both, then evaluate them for $E = 7.1 \times 10^{13}$ J. What is the physical meaning of R^*?

Task 2 Reading the data

15 min

(a) Compute $Q = R^5/t^2$ for $t = 0.10$, 0.94, 15.0 and 62.0 ms. What do you notice about the first value? What about the last?

(b) Use the points $t = 0.24$ ms and $t = 62.0$ ms to estimate the slope of the log–log plot. Compare it with $2/5$.

(c) Using Taylor's values $Q = 6.67 \times 10^{13}\,\mathrm{m^5\,s^{-2}}$ and $K = 0.856$, calculate E in joules and in kilotons of TNT.

(d) *A referee's eye.* Taylor's paper fits the line $\frac{5}{2}\log_{10} R - \log_{10} t = 11.915$, with R in cm and t in s. Find the value of Q in SI units that this implies, and compare it with the value in (c). Comment on what you find.

Task 3 Testing the assumptions

15 min

For a strong shock in a perfect gas, the conditions just behind the front are $\rho_2 = \frac{\gamma+1}{\gamma-1}\rho$ and $p_2 = \frac{2}{\gamma+1}\rho U^2$, where U is the speed of the front. (You will derive these in Competition Corner.)

(a) Show from Eq. (1) that $U = \dfrac{2R}{5t}$.

(b) For $\gamma = 1.4$, complete this table. Use $T_2 = p_2/(\rho_2 r)$ with $r = 287\,\mathrm{J\,kg^{-1}\,K^{-1}}$.

t (ms)	R (m)	U (m/s)	Mach number U/c	p_2/p_0	T_2 (K)
0.94	36.3				
15.0	106.5				
62.0	185.0				

(c) Is the strong-shock assumption justified throughout Table 1? Now look at your T_2 values. Why is it surprising that a single constant γ works?

(d) Take the mass of the device to be about 5 tonnes. At what radius does the mass of swept-up air equal this value? Use your answer to explain the earliest point in Table 1.

Task 4 How large is the error? 5 min

(a) Show that the maximum relative error in E is $\dfrac{\delta E}{E} = 5\dfrac{\delta R}{R} + 2\dfrac{\delta t}{t}$. Evaluate it for a 2% error in R and a 1% error in t.

(b) Which matters more here, the measurement error in (a) or the choice between $K = 0.856$ ($\gamma = 1.4$) and $K = 1.167$ ($\gamma = 1.3$)? Justify your answer with numbers.

Code Lab 代码实验 课内·15 min

这是第一个 Code Lab，此时还没有学 Python（C0 在下下次课），所以代码已经写好，只需运行、修改参数并解释结果。文件在 `code/P1_code_lab.py`。

```
# Unit P1 Code Lab: fitting Taylor's blast-wave data
import numpy as np
import matplotlib.pyplot as plt

# Taylor (1950), Table 1: time in ms, radius in m
t_ms = np.array([0.10, 0.24, 0.38, 0.52, 0.66, 0.80, 0.94, 1.08, 1.22,
                 1.36, 1.50, 1.65, 1.79, 1.93, 3.26, 3.53, 3.80, 4.07,
                 4.34, 4.61, 15.0, 25.0, 34.0, 53.0, 62.0])
R = np.array([11.1, 19.9, 25.4, 28.8, 31.9, 34.2, 36.3, 38.9, 41.0,
              42.8, 44.4, 46.0, 46.9, 48.7, 59.0, 61.1, 62.9, 64.3,
              65.6, 67.3, 106.5, 130.0, 145.0, 175.0, 185.0])
t = t_ms / 1000            # convert ms to s

# ---- Things to change ----
t_min = 0.2                # ignore points earlier than this (ms)
K = 0.856                  # K = 0.856 for gamma = 1.4; K = 1.167 for gamma = 1.3
rho = 1.25                 # density of air, kg/m^3
# --------------------------

keep = t_ms >= t_min
log_t, log_R = np.log10(t[keep]), np.log10(R[keep])

# 1. Let the data choose the slope
slope, intercept = np.polyfit(log_t, log_R, 1)
print(f"points used: {keep.sum()}")
print(f"free fit: slope = {slope:.3f}")

# 2. Fix the slope at 2/5 and find Q = R^5 / t^2
log_c = np.mean(log_R - 0.4 * log_t)
Q = 10 ** (5 * log_c)
E = K * rho * Q
print(f"Q = {Q:.2e} m^5/s^2")
print(f"E = {E:.2e} J = {E / 4.184e12:.1f} kt of TNT")

# 3. Plot the data and both lines
plt.loglog(t_ms, R, "k+", label="Taylor (1950)")
line_t = np.logspace(-4, -1, 50)
plt.loglog(line_t * 1000, 10**log_c * line_t**0.4, label="slope 2/5")
plt.loglog(line_t * 1000, 10**intercept * line_t**slope, "--", label="free fit")
plt.xlabel("t (ms)")
plt.ylabel("R (m)")
plt.legend()
plt.savefig("taylor_fit.png", dpi=150)
```

```
plt.show()
```

1. Run the program and record the slope and the energy. Look at the plot: do the two lines differ much?
2. Set `t_min` to `0.0`, `0.5` and `10.0` in turn. Record the slope and the energy each time. Which data points have the greatest influence?
3. Set `K = 1.167`. What is the energy now?
4. The program calculates E only after fixing the slope at $2/5$. Explain in English why the free-fit line cannot be used in Eq. (2).

Competition Corner 竞赛角

课后

Problem 1 A thin-shell model of a blast wave (12 points)

Taylor had to integrate the equations numerically to find K. In this problem you will find a good approximation by hand. A strong spherical shock of radius $R(t)$ and speed $U = \dot{R}$ moves into a still perfect gas of density ρ and negligible pressure. Behind the front, the gas has density ρ_2, pressure p_2 and velocity u in the laboratory frame.

A. (3 points) Work in the frame of the shock. Write down the conservation of mass, momentum and energy across the front, using the specific enthalpy $h = \dfrac{\gamma p}{(\gamma-1)\rho}$. Hence show that

$$\frac{\rho_2}{\rho} = \frac{\gamma+1}{\gamma-1}, \qquad p_2 = \frac{2}{\gamma+1}\rho U^2, \qquad u = \frac{2U}{\gamma+1}.$$

B. (1 point) Show that, just behind the front, the kinetic energy per unit mass equals the internal energy per unit mass.

C. (3 points) Figure 1b suggests a simple model. All the swept-up mass $M = \frac{4}{3}\pi R^3 \rho$ moves with speed u in a thin shell at radius R, and the interior has uniform pressure p_c. Given that $R \propto t^{2/5}$, apply Newton's second law to a small sector of the shell and show that $p_c = \frac{1}{2}p_2$.

D. (3 points) Take the energy to be the kinetic energy of the shell plus the internal energy of the interior. Show that $E = K_{\mathrm{model}}\,\rho R^5/t^2$ with

$$K_{\mathrm{model}} = \frac{16\pi}{75}\left[\frac{2}{(\gamma+1)^2} + \frac{1}{(\gamma+1)(\gamma-1)}\right].$$

Evaluate K_{model} for $\gamma = 1.4$ and compare it with the exact value 0.851.

E. (2 points) What fraction of the energy is kinetic in this model? The exact value is 22%. By what percentage does the model's error in K change the predicted radius R at a given time?

Problem 2 A blast wave in the Galaxy (8 points)

A supernova releases $E = 1.0 \times 10^{44}\,\mathrm{J}$ into interstellar gas of density $\rho = 1.67 \times 10^{-21}\,\mathrm{kg\,m^{-3}}$ (about one hydrogen atom per cubic centimetre). The gas has $\gamma = 5/3$, for which $S = 1.15$. Use $1\,\mathrm{pc} = 3.09 \times 10^{16}\,\mathrm{m}$, $1\,\mathrm{yr} = 3.16 \times 10^{7}\,\mathrm{s}$, $M_\odot = 1.99 \times 10^{30}\,\mathrm{kg}$, $m_p = 1.67 \times 10^{-27}\,\mathrm{kg}$ and $k_B = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$.

A. (3 points) The ejected mass is $1.4 M_\odot$. If all the energy is initially kinetic, find the ejecta speed. At what radius, and after how many years, has the ejecta swept up its own mass of interstellar gas? Why is Eq. (1) not valid before this?

B. (2 points) Find R in parsecs and U in km/s, 1000 years after the explosion.

C. (3 points) Show that the temperature just behind the shock is $T_2 = \dfrac{3\mu m_p U^2}{16 k_B}$, where μ is the mean particle mass in units of m_p. Evaluate T_2 for $\mu = 0.6$. In which part of the electromagnetic spectrum should such gas glow?

Output 学术产出

A：课内 15 min B：课后

A. Speak: a two-minute explanation

Without reading from notes, explain to an imaginary examiner how Taylor could estimate the energy of an explosion from photographs. Cover the assumptions, the scaling law, the data and one limitation. These phrases may help:

The key question is ... *If we assume that ..., then only three quantities matter.* *It follows that ...* *On a log–log plot, this appears as ...* *However, this argument cannot tell us ...* *The main limitation is that ...*

B. Write: your first abstract (120–150 words)

Write an abstract for a short report on your own work in Task 2, Task 3 and the Code Lab. An abstract usually makes four moves. Do not copy sentences from the Main Text.

Move	作用	Useful language
Background	为什么值得研究	The energy of ...can be inferred from ...
Method	做了什么	We fit ...to ...; We test whether ...
Result	得到的数值	We find a slope of ...; This gives ...
Implication	结果的意义和局限	This agrees to within ...; The model fails when ...

Answers 参考答案

Warm-up

1. $16\,000/340 \approx 47$ s. Fermi reported about 40 s. A strong blast travels faster than sound, particularly in its early stage, so the estimate is a little too long.
   Fermi 还丢下纸片，由纸片约 2.5 m 的位移估算出约 10 kt。他如何由位移得到这个数字，至今没有完整的解释（Katz, *Fermi at Trinity*, 2021）。
2. $17\text{ kt} = 7.1\times10^{13}\text{ J} = 2.0\times10^{7}\text{ kWh}$. The city uses $2\times10^{11}/8760 = 2.3\times10^{7}$ kWh per hour, so the energy lasts about 50 minutes.
3. Open discussion. The energy, air density, time and radius matter. The air temperature, the ambient pressure and the details of the device might be ignored, but this needs justification.

Word Work

1. negligible 2. break down 3. Substituting 4. unity 5. exceeds 6. sensitive 7. yields 8. dispense with

Reading Check

1. False (¶2). The theory dates from 1941, four years before the test.
2. False (¶6). It gives the exponent, but S requires integrating the flow equations.
3. False (¶8). It counts only the energy not radiated away beyond the fireball.
4. False (¶9). It falls below the line, because the device and radiation still matter at that stage.
5. The point-source assumption removes the mass and size of the device. The strong-shock assumption removes the ambient pressure.
6. A scientist looks for measurable quantities in a picture, not just its visual impact. Here the time stamp and scale turn it into a measurement.
7. They suggest that dimensional analysis was the whole of his method, when in fact he also solved the equations of flow to obtain the constant.
8. The law survives even though one of its assumptions, a constant γ, is certainly untrue at such high temperatures.
9. (a) ¶9; (b) ¶3; (c) ¶8; (d) ¶10.
   ¶7 把理论与照片数据比较，但照片本身就是用来定出 E 的，不算独立检验。(c) 的独立测量是 ¶8 的放射化学结果。
10. Any two of these: (i) The explosion is spherically symmetric, although the fireball soon reached the ground. Taylor noted that the upper part still expanded as a nearly perfect hemisphere. (ii) The luminous edge marks the shock front. At late times the shock moves ahead of the luminous region. (iii) The air density is uniform and equal to its value at ground level.

Language Focus

A. (1) R is proportional to t to the power of two-fifths. (2) E equals K rho R to the fifth over t squared. (3) p two is much greater than p nought. (4) Delta E over E is approximately five delta R over R. (5) E is of the order of ten to the fourteen joules.

B.

1. proportional **to** $t^{2/5}$.

2. The relation is reversed: **The ambient pressure** is negligible compared with **the pressure behind the front**.
3. is **of** the order of 10^{14} J.
4. makes the energy **32 times** larger ($2^5 = 32$).
5. thirty orders **of magnitude** more energy.

C.（参考译文）

1. The radius of the blast wave grows as the two-fifths power of time.
2. The pressure behind the shock is about 80 times the ambient pressure, so the latter is negligible.
3. This estimate agrees with the modern value to within about 30%.
4. A 2% error in the radius leads to a 10% error in the energy.

Work It Out

Task 1. (a) Matching M, L and T gives $a + b = 0$, $2a - 3b = 1$ and $-2a + c = 0$, so $a = \frac{1}{5}$, $b = -\frac{1}{5}$, $c = \frac{2}{5}$.

(b) $\Pi = \dfrac{p_0\, t^{6/5}}{E^{2/5}\rho^{3/5}}$. There are now five quantities (R, E, ρ, t, p_0) and three dimensions, so there are two independent dimensionless groups. One can only say that the first is a function of the second. In the strong-shock limit $\Pi \to 0$, and $f(0) = S$.

(c) $t^* = E^{1/3}\rho^{1/2}p_0^{-5/6} \approx 3.1\,\mathrm{s}$ and $R^* = (E/p_0)^{1/3} \approx 890\,\mathrm{m}$. R^* is the radius at which the energy per unit volume, $\sim E/R^3$, falls to the ambient pressure.

此时前沿速度 $2R^*/5t^* \approx 115\,\mathrm{m/s}$，已低于声速，说明强激波近似在此之前就已失效。按式 (1) 估算，$M = 3$ 约在 $t \approx 0.085\,\mathrm{s}$、$R \approx 220\,\mathrm{m}$，$M = 2$ 约在 $t \approx 0.17\,\mathrm{s}$、$R \approx 285\,\mathrm{m}$，所以数据（到 185 m）恰好在失效之前结束。

Task 2. (a) $Q = 1.69\times10^{13}$, 7.13×10^{13}, 6.09×10^{13}, $5.64\times10^{13}\ \mathrm{m^5\,s^{-2}}$. The first value is about a quarter of the others, so the earliest point lies outside the regime of the law. The value at late times drifts slightly downward, an early sign that the shock is weakening.

(b) slope $= \dfrac{\log_{10}(185/19.9)}{\log_{10}(62.0/0.24)} = \dfrac{0.968}{2.412} = 0.401$, very close to 0.4.

(c) $E = 0.856\times1.25\times6.67\times10^{13} = 7.14\times10^{13}\,\mathrm{J} \approx 17\,\mathrm{kt}$.

Taylor 写作 16,800 吨，差别来自他用的换算值 $1\,\mathrm{t} = 4.25\times10^{16}\,\mathrm{erg}$，现代定义是 $4.184\times10^{16}\,\mathrm{erg}$。

(d) $R^5/t^2 = 10^{2\times11.915} = 6.76\times10^{23}\,\mathrm{cm^5\,s^{-2}} = 6.76\times10^{13}\,\mathrm{m^5\,s^{-2}}$. This differs from the printed 6.67×10^{13} by about 1.3%. The difference is far smaller than the other uncertainties, but a careful reader should notice it.

原文的式 (1) 和式 (2) 确实有这一差异，后续的能量用的是 6.67。这里不必判断是哪一个错了，重点是养成核对的习惯。

Task 3. (a) $R = At^{2/5}$ gives $U = \dot R = \frac{2}{5}At^{-3/5} = \dfrac{2R}{5t}$.

(b)

t (ms)	R (m)	U (m/s)	U/c	p_2/p_0	T_2 (K)
0.94	36.3	1.54×10^4	45	2450	1.2×10^5
15.0	106.5	2.84×10^3	8.4	83	3.9×10^3
62.0	185.0	1.19×10^3	3.5	15	6.9×10^2

(c) Yes: even at the last point $p_2/p_0 \approx 15$, although the approximation is getting weaker there. The temperatures range over more than two orders of magnitude. Air molecules would be vibrationally excited, dissociated and ionised at various stages, so γ cannot truly be constant. Taylor himself called this assump-

tion “certainly not true”.

表中早期的 T_2 是理想气体公式的结果，只表示量级。实际空气在如此高温下会电离并强烈辐射。

(d) $\tfrac{4}{3}\pi R^3\rho = 5000\,\mathrm{kg}$ gives $R \approx 10\,\mathrm{m}$. The earliest point, $R = 11.1\,\mathrm{m}$, is at about this radius, so the point-source assumption does not yet hold and the point should not be expected to lie on the line. Taylor’s line begins at $R = 20\,\mathrm{m}$.

Task 4. (a) $\ln E = \ln(K\rho) + 5\ln R - 2\ln t$. Differentiating and adding the magnitudes gives the result. $\delta E/E = 5(2\%) + 2(1\%) = 12\%$.

(b) $1.167/0.856 = 1.36$, a 36% change, which is about three times the measurement error. The largest uncertainty therefore lies in the physics of hot air, not in the ruler or the clock.

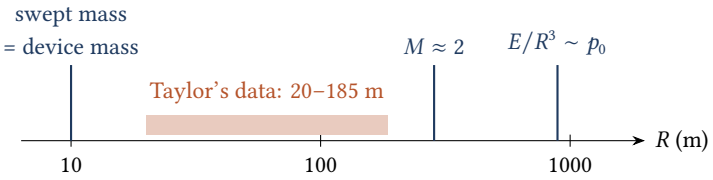


Figure 3. The window in which Eq. (1) applies, on a logarithmic scale of radius.

Code Lab

`t_min` (ms)	Points used	Free-fit slope	Q (m^5 s^{-2})	E (kt)
0.0	25	0.406	6.42×10^{13}	16.4
0.2	24	0.390	6.79×10^{13}	17.4
0.5	22	0.386	6.84×10^{13}	17.5
10.0	5	0.391	5.81×10^{13}	14.9

1. Slope 0.390 and $E = 17.4$ kt. Both lines pass close to the data, and they separate visibly only at the ends.
2. The single earliest point visibly shifts the slope, because it lies far to the left and below the line. The late-time points alone give a lower energy, because Q falls as the shock weakens.
3. $E = 23.7$ kt with the same data.
4. Eq. (2) requires R^5/t^2 to be constant, which is true only if the slope is exactly $2/5$. On the free-fit line this quantity varies with time, so no single value of E can be read from it.

Competition Corner

Problem 1.

A. In the shock frame, gas enters at speed U and leaves at speed $w = U - u$. With negligible upstream pressure,

$$\rho U = \rho_2 w, \qquad \rho U^2 = p_2 + \rho_2 w^2, \qquad \tfrac{1}{2}U^2 = \frac{\gamma p_2}{(\gamma-1)\rho_2} + \tfrac{1}{2}w^2.$$

Let $n = \rho_2/\rho$, so $w = U/n$. Momentum gives $p_2 = \rho U^2(1 - 1/n)$. Energy then gives $\tfrac{1}{2}(1 - 1/n^2) = \tfrac{\gamma}{\gamma-1}\tfrac{1}{n}(1 - 1/n)$, which reduces to $\tfrac{n+1}{2} = \tfrac{\gamma}{\gamma-1}$. So $n = \tfrac{\gamma+1}{\gamma-1}$, and the expressions for p_2 and $u = U(1 - 1/n)$ follow.

B. $e = \dfrac{p_2}{(\gamma-1)\rho_2} = \dfrac{2U^2}{(\gamma+1)^2} = \dfrac{1}{2}u^2$.

C. $R \propto t^{2/5}$ means $U \propto R^{-3/2}$, so $Mu \propto R^3 \cdot R^{-3/2} = R^{3/2}$. Then

$$\frac{d(Mu)}{dt} = \frac{3}{2}\frac{Mu}{R}U = 4\pi R^2 \frac{\rho U^2}{\gamma+1}.$$

Setting this equal to the force $4\pi R^2 p_c$ gives $p_c = \dfrac{\rho U^2}{\gamma+1} = \dfrac{1}{2}p_2$.

精确解中心压强约为 $0.37\,p_2$，壳层附近压强更高，所以取平均值 $\frac{1}{2}p_2$ 是合理的。

D. With $V = \frac{4}{3}\pi R^3$: $\frac{1}{2}Mu^2 = \dfrac{2\rho V U^2}{(\gamma+1)^2}$ and $\dfrac{p_c V}{\gamma-1} = \dfrac{\rho V U^2}{(\gamma+1)(\gamma-1)}$. Substituting $U^2 = \frac{4}{25}R^2/t^2$ gives the stated formula. For $\gamma = 1.4$, $K_{\text{model}} = 0.670 \times (0.347 + 1.042) = 0.931$, which is about 9% above the exact value.

E. Kinetic fraction $= 0.347/1.389 = 25\%$. Since $R \propto K^{-1/5}$, a 9% error in K changes R by only about 2%. ($S_{\text{model}} = 1.014$, against the exact 1.033.)

Problem 2.

A. $v = \sqrt{2E/M} = 8.5 \times 10^3$ km/s. Setting $\frac{4}{3}\pi R^3 \rho = 1.4 M_\odot$ gives $R = 7.4 \times 10^{16}$ m $= 2.4$ pc, and $t = R/v \approx 2.8 \times 10^2$ yr. Before this, the ejecta dominates and expands almost freely, so the point-source assumption fails.

B. $R = 1.15\,(Et^2/\rho)^{1/5} = 1.6 \times 10^{17}$ m $= 5.3$ pc and $U = 2R/5t = 2.1 \times 10^3$ km/s.

C. For $\gamma = \frac{5}{3}$: $\rho_2 = 4\rho$ and $p_2 = \frac{3}{4}\rho U^2$, so $\dfrac{k_B T_2}{\mu m_p} = \dfrac{p_2}{\rho_2} = \dfrac{3U^2}{16}$. Hence $T_2 \approx 5.9 \times 10^7$ K, or $k_B T_2 \approx 5$ keV. Such gas glows in X-rays, which is how remnants of this kind are observed.

Output B: a model abstract

> The energy of an intense explosion can be inferred from the growth of its fireball, provided that the blast wave is strong and the source is effectively a point. We test this idea using the radius–time measurements of the 1945 Trinity explosion published by Taylor (1950), excluding the earliest point. A least-squares fit to the log–log data gives a slope of 0.39, close to the predicted value of 2/5. Fixing the slope at 2/5 yields an energy of about 17 kilotons of TNT. We find that the shock remains strong throughout most of the data, with a Mach number that falls from about 100 to 3.5. The largest uncertainty lies in the effective value of γ, which changes the result by about 36%. Our estimate agrees with the modern yield to within about 30%.

字数约 130。注意四个步骤：背景（第 1 句）、方法（第 2 句）、结果（第 3–5 句）、意义与局限（第 6–7 句）。